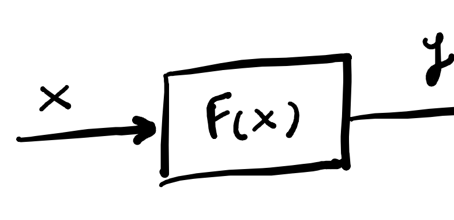
INTRO SCHEMI A BLOCCHI

▪ RAPPRESENTAZIONE e ANALISI dei SISTEMI

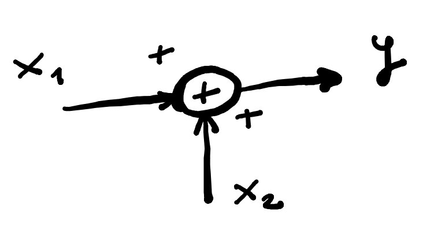
- DAI CIRCUITI AI SISTEMI
- RAPPRESENTAZIONE dei SISTEMI:
  - i) SCHEMI a BLOCCHI → UTILIZZO di BLOCCHI IDEALI per la RAPPRESENTAZIONE del COMPORTAMENTO dei SISTEMI
  - ii) DALLO SCHEMA a BLOCCHI alle RELAZIONE IN/OUT
  - iii) DALLA RELAZIONE IN/OUT ALLO SCHEMA a BLOCCHI
- SISTEMI in CASCATA e in PARALLELO
- DIFFERENZA TRA SISTEMI IDEALI e REALI

• SCHEMI A BLOCCHI → BLOCCHI IDEALI



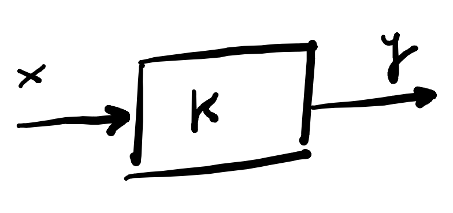
$y = F(x)$

SISTEMA e RELAZIONE IN/OUT: LA RELAZIONE può ESSERE una FUNZIONE ANCHE COMPLESSA o in GENERALE una TRASFORMAZ.



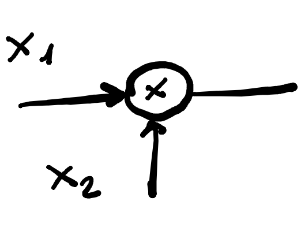
$y = x_1 + x_2$

SOMMATORE



$y = Kx$

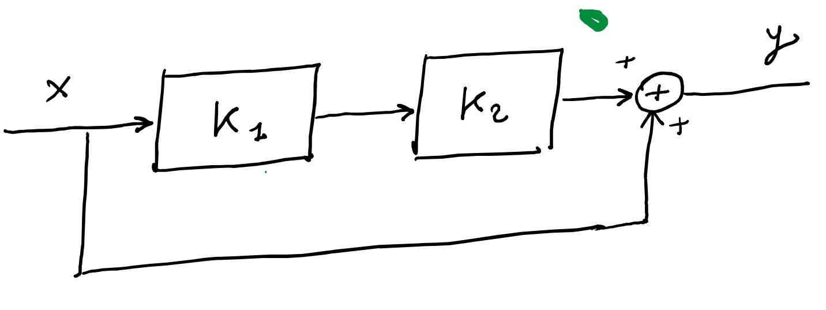
BLOCCO GUADAGNO

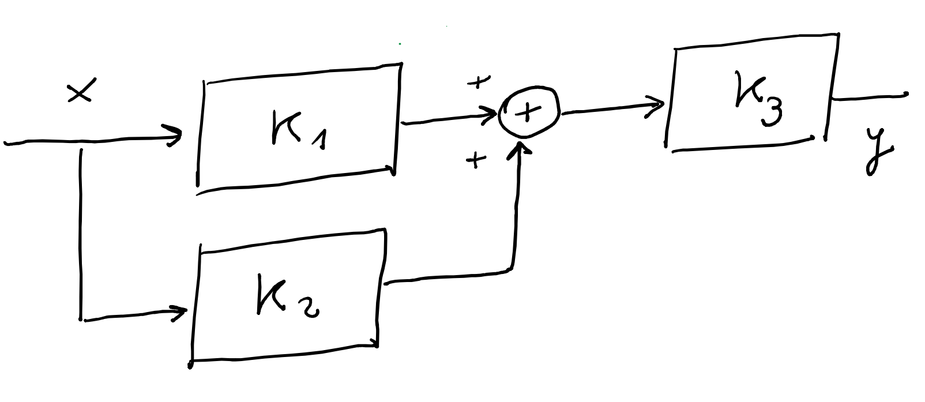


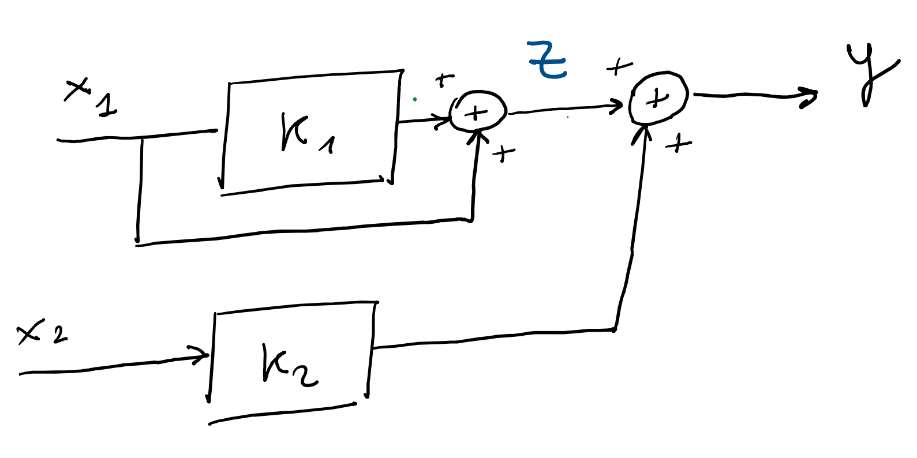
$y = x_1 x_2$

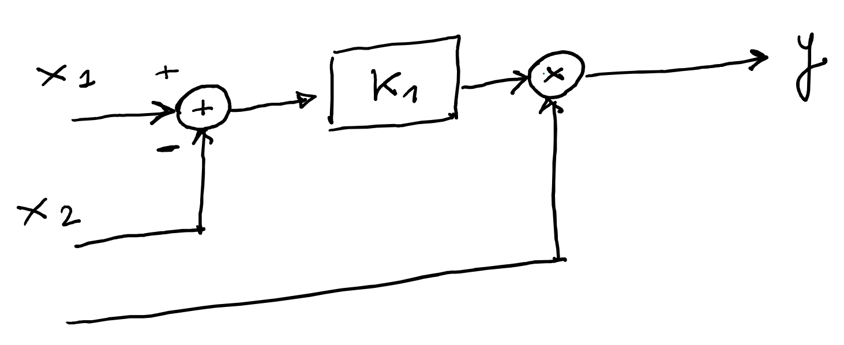
BLOCCO MOLTIPLICATORE

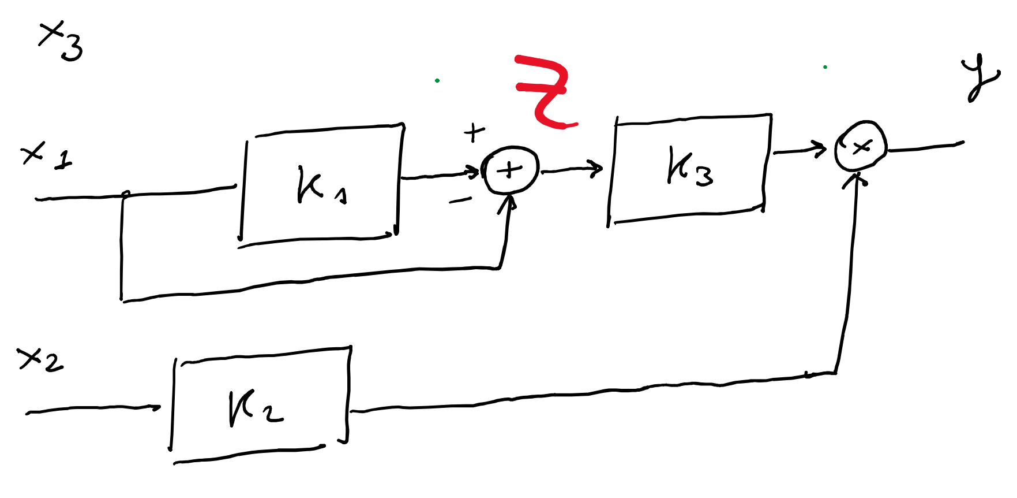
• ESEMPI USO SCHEMA A BLOCCHI: DA SCHEMA A BLOCCHI ALLA RELAZIONE IN/OUT





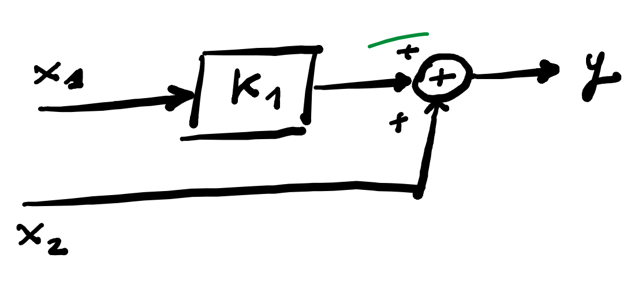






• DALLA RELAZIONE IN/OUT ALLO SCHEMA a BLOCCHI:

i) $y = K_1 x_1 + x_2$ →



ii) $y = (K_1 - 1)x_1 + K_2 x_2$

iii) $y = K_1(x_1 - x_2) + x_3$

iv) $y = K_1(x_1 - x_2)x_3$

v) $y = K_1(x_1 x_2) + x_3$